

$$u(x_1, x_2) = K x_1^\alpha x_2^{1-\alpha}$$

$$s.t. \quad P_1 x_1 + P_2 x_2 = W$$

Setting up Lagrangian:

$$L(x_1, x_2; \lambda) = K x_1^\alpha x_2^{1-\alpha} + \lambda (W - P_1 x_1 - P_2 x_2)$$

FOCs:

$$\frac{\partial L}{\partial x_1} = K \alpha x_1^{\alpha-1} x_2^{1-\alpha} - \lambda P_1 = 0 \quad (i)$$

$$\frac{\partial L}{\partial x_2} = K x_1^\alpha (1-\alpha) x_2^{-\alpha} - \lambda P_2 = 0 \quad (ii)$$

$$\frac{\partial L}{\partial \lambda} = W - P_1 x_1 - P_2 x_2 = 0 \quad (iii)$$

From (i)

$$\lambda P_1 = K \alpha x_1^{\alpha-1} x_2^{1-\alpha} \quad (iv)$$

from (ii)

$$\lambda P_2 = K (1-\alpha) x_1^\alpha x_2^{-\alpha} \quad (v)$$

$$(iii) \div (iv)$$

$$\frac{P_1}{P_2} = \frac{\alpha}{(1-\alpha)} \cdot \frac{x_2}{x_1} \quad (v)$$

From (v) $w = \frac{P_1 x_1}{P_2 x_2} \left(1 + \frac{1}{b-1} \right) \quad (vi)$

$$P_1 x_1 = \frac{\alpha}{(1-\alpha)} P_2 x_2 \quad (vii)$$

and, $P_2 x_2 = \frac{(1-\alpha)}{\alpha} P_1 x_1 \quad (viii)$

Substituting (viii) into budget constraint:

$$P_1 x_1 + \frac{(1-\alpha)}{\alpha} P_1 x_1 = w$$

$$\Rightarrow P_1 x_1 \left(1 + \frac{1-\alpha}{\alpha} \right) = w$$

$$\Rightarrow \frac{1}{\alpha} w P_1 x_1 = w$$

$$\therefore x_1 = \frac{w}{P_1}$$

Substituting (ii) into budget constraint (iii)

$$\frac{\alpha}{(1-\alpha)} P_2 x_2 + P_2 x_2 = w \quad \frac{5}{10} \quad \frac{5}{(5-1)} = \frac{5}{4}$$

$$\Rightarrow \left(\frac{\alpha}{1-\alpha} + 1 \right) P_2 x_2 = w \quad (v) \text{ now}$$

$$\Rightarrow \frac{1}{(1-\alpha)} \cdot P_2 x_2 = w \quad \frac{5}{(5-1)} = 1.25$$

$$\Rightarrow x_2 = \frac{w(1-\alpha)}{P_2} \quad \frac{1.25 \cdot (5-1)}{5} = 1 \quad \text{Ans}$$

$\therefore$ Walrasian demand for good 1, $x_1(P_1, w) = \frac{2w}{P_1}$

$\therefore$ Walrasian demand for good 2, $x_2(P_2, w) = \frac{w(1-\alpha)}{P_2}$